

CARLISLE, United Kingdom

WMO: 032200

Lat: 54.9344N

Lon: 2.9636W

Elev: 27

StdP: 101.00

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.4	(c) -2.6	(d) -5.7	(e) 2.3	(f) -1.5	(g) -4.4	(h) 2.6	(i) -0.3	(j) 15.8	(k) 8.2	(l) 14.4	(m) 8.7	(n) 0.5	(o) 80	(p) 0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.5	(c) 23.8	(d) 17.8	(e) 21.9	(f) 16.8	(g) 20.2	(h) 16.0	(i) 18.8	(j) 22.5	(k) 17.7	(l) 20.7	(m) 16.7	(n) 19.4	(o) 2.6	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.5	20.1	16.4	11.7	19.1	15.6	11.1	18.2	53.3	22.6	49.9	20.7	47.1	19.5	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.4	8.9	-6.5	27.9	2.0	2.4	-8.0	29.6	-9.1	31.0	-10.2	32.4	-11.7	34.1
			-6.5	21.1	2.1	1.8	-8.0	22.4	-9.3	23.4	-10.5	24.4	-12.0	25.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	4.6	4.9	6.1	8.3	11.2	13.9	15.7	15.5	13.4	10.4	7.0	4.7
	DBStd	4.95	3.22	2.96	2.72	2.65	2.74	2.26	2.19	2.09	2.28	2.84	3.31	3.95
	HDD10.0	811	167	144	123	62	18	1	0	0	2	29	97	167
	HDD18.3	3176	424	377	379	299	223	135	88	93	149	246	339	423
	CDD10.0	687	1	1	2	13	54	117	176	169	103	42	8	2
	CDD18.3	12	0	0	0	0	1	1	6	4	1	0	0	0
	CDH23.3	84	0	0	0	0	3	14	40	24	2	0	0	0
	CDH26.7	12	0	0	0	0	0	2	7	3	0	0	0	0
(14) Wind	WSAvg	2.9	3.8	4.0	3.6	3.2	2.7	2.5	2.2	2.0	2.3	2.7	2.8	3.0
(15) Precipitation	PrecAvg	776	70	50	57	48	54	59	68	75	78	78	74	70
	PrecMax	1023	134	175	113	112	107	165	188	158	177	213	175	130
	PrecMin	572	5	4	19	4	18	6	20	6	12	15	24	12
	PrecStd	111	33	35	24	23	23	33	33	38	35	37	35	31
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.0	15.1	19.6	23.3	25.2	27.2	26.5	22.4	18.2	15.1	13.5
		MCWB	10.8	10.2	10.4	14.0	16.5	18.7	18.9	19.1	17.1	14.4	13.1	11.6
	2%	DB	11.1	11.2	12.5	16.0	20.3	22.2	23.9	22.6	20.1	16.7	13.7	12.3
		MCWB	9.6	9.4	9.7	11.4	14.9	16.7	18.0	17.4	15.9	14.0	12.0	10.8
	5%	DB	10.2	10.4	11.2	14.2	17.9	20.0	21.8	20.6	18.5	15.7	12.7	11.1
		MCWB	8.8	8.8	9.0	10.5	13.6	15.4	17.2	16.5	15.1	13.4	11.1	9.7
	10%	DB	9.3	9.4	10.2	12.8	15.9	18.3	19.9	19.3	17.3	14.7	11.8	10.1
		MCWB	8.0	7.9	8.3	9.7	12.1	14.5	16.0	15.8	14.3	12.8	10.4	8.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.0	11.7	14.1	17.6	19.6	21.1	19.7	18.5	15.6	13.6	12.1
		MCDB	12.1	12.0	14.1	17.8	22.0	23.5	23.8	24.0	21.2	16.9	14.7	13.2
	2%	WB	10.1	10.0	10.4	12.2	15.4	17.5	19.0	18.4	16.9	14.7	12.4	11.0
		MCDB	11.0	11.0	11.9	14.9	19.6	21.2	22.9	21.5	19.0	16.1	13.4	12.1
	5%	WB	9.1	9.1	9.5	11.1	14.0	16.1	17.7	17.4	15.7	13.9	11.5	10.0
		MCDB	10.0	10.2	10.9	13.6	17.0	19.2	20.9	19.8	17.6	15.5	12.5	11.1
	10%	WB	8.1	8.1	8.6	10.1	12.8	14.9	16.6	16.5	14.8	12.9	10.5	8.9
		MCDB	9.2	9.3	10.0	12.4	15.4	17.6	19.3	18.6	16.8	14.4	11.6	10.1
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.4	6.3	7.4	8.0	7.6	7.5	7.3	7.1	6.2	5.4	5.1
		MCDBR	5.1	5.7	7.2	9.6	11.2	10.9	10.9	9.8	8.8	6.5	5.1	5.4
		MCWBR	4.8	4.7	5.0	6.1	6.7	6.5	6.1	5.6	5.4	4.7	4.6	5.1
	5% WB	MCDBR	5.1	5.2	6.6	8.8	9.9	9.5	8.4	7.4	5.8	5.3	5.3	5.6
		MCWBR	5.0	4.7	5.0	5.9	6.5	6.4	5.9	5.3	5.3	4.9	4.9	5.4
(40) Clear-Sky Solar Irradiance	taub		0.328	0.339	0.368	0.389	0.391	0.389	0.389	0.377	0.362	0.346	0.322	0.312
	taud		2.247	2.328	2.275	2.250	2.291	2.327	2.331	2.377	2.422	2.410	2.326	2.228
	Ebn at Noon		586	735	795	832	852	857	849	838	801	720	597	513
	Edh at Noon		60	78	102	119	121	118	116	104	88	71	54	49
(44) All-Sky Solar Radiation	RadAvg		0.55	1.21	2.29	3.67	4.78	4.74	4.58	3.69	2.63	1.43	0.70	0.42
	RadStd		0.07	0.13	0.22	0.38	0.37	0.47	0.49	0.28	0.23	0.17	0.07	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page